Cross-Modal Oriented Object Detection of UAV Aerial Images Based on Image Feature

Huiying Wang, Chunping Wang, Qiang Fu, Dongdong Zhang, Renke Kou, Ying Yu, and Jian Song

***Abstract*— Arbitrary-oriented object detection is vital for improving unmanned aerial vehicle (UAV) sensing and has promising applications. However, challenges persist in detecting objects under extreme conditions like low-illumination and strong occlusion. Cross-modal feature fusion enhances detection in complex environments but current methods do not adequately learn the features of each modality for the current environment, resulting in degraded performance. To tackle this, we propose the cross-modal aerial remote sensing image object detection (CRSIOD) network that effectively learns diverse sensor image features to capture distinct scenarios and target characteristics. First, we design an illumination perception module to guide the object detection network in performing various feature processing tasks. Second, to leverage the respective advantages of two modalities and mitigate their negative impacts, we introduce an uncertainty-aware module to quantify the uncertainties present in each modality as weights to motivate the network to learn in a direction favorable for optimal object detection. Moreover, in the object detection network, we design a two-stream backbone network based on the attention mechanism to enhance the learning of difficult samples, utilize the cross-modality attentive feature fusion (CMAFF) module to fully extract the shared and complementary features between the two modalities, and design a three-branch feature enhancement network to enhance the learning of the three modal features separately. Finally, to optimize detection results, we design light perception nonmaximum suppression (LP-NMS) and improve the horizontal detection head to a rotating one to preserve object orientation. We evaluate the proposed method CRSIOD on the Drone Vehicle dataset of public UAV aerial images. Compared with the existing commonly used methods, CRSIOD achieves state-of-the-art detection performance.**

***Index Terms*— Feature fusion, multimodal aerial images, object detection, oriented object.**

I. Introduction

Unmanned aerial vehicle (UAV) has the advantages of high flexibility, ease of operate, tiny size, and low cost, which can make up for the shortcomings of satellite and manned aerial remote sensing technology, and spawn more diversified application scenarios. UAV-based vehicle detection plays an important role in many civilian and military applications, such as road vehicle monitoring, intelligent traffic management, missing vehicle search, disaster relief, and rescue [1], [2], [3], [4], [5], [6].

The camera-equipped drones can be airborne for detection without road restrictions, with advantages such as speed, high freedom, and wide field of view. However, because of extreme conditions, such as highly complex backgrounds and very low light levels, aerial image-based object detection remains an active and challenging task in computer vision [7].

Traditional machine-learning schemes have made significant advancements in object detection using the encoding of handcrafted features, such as scale-invariant feature transform (SIFT) [8], Haar [9], and histogram of oriented gradient (HOG) [10]. However, most of these algorithms rely on expert prior knowledge, which has certain limitations and cannot cope with the complexity and variability of the background as well as the multimodality and variability of the target. In recent years, along with the rapid development of deep learning technology, object detection methods based on convolutional neural networks (CNNs) have been widely used because of their strong learning ability and high detection accuracy. Currently, the mainstream deep learning object detection methods are mainly divided into two categories: two-stage network [11], [12], [13], [14], [15], [16] and single-stage network [17], [18], [19], [20]. The two-stage detector, also called the region-based detector, divides the object detection task into two stages: localization and recognition. Usually, two-stage network detection is more accurate but computationally intensive and time consuming. Conversely, single-stage detector do not have a separate stage for proposal generation, but they work faster than two-stage detector, which is very important for industry.

Recently, some object detection algorithms for aerial images have emerged [21], [22], [23], [24], [25], [26]. For targets in complex scenes, U^2D^2Net [27] proposed to unify the defogging and noise reduction tasks by introducing a regional similarity fusion module to improve the clarity of visible light images in low visibility. However, applying visible images alone in low light or no-light scenes at night cannot effectively monitor targets. Since then, it has been found that adopting multiple sensor sources by using multimodal fusion technology can improve the performance in complex scenarios [28]. The advantage of thermal images being less susceptible to illumination has received a lot of attention from researchers. Aiming at the characteristics of thermal targets, Deep-IRTarget [29] combines frequency- and space-domain features to form a dual-domain feature, which improves the accuracy of thermal object detection. Zhang et al. [30] proposed to convert the features extracted by a pretrained

Manuscript received 13 November 2023; revised 27 December 2023; accepted 6 February 2024. Date of publication 20 February 2024; date of current version 28 February 2024. This work was supported by Hainan Provincial Natural Science Foundation of China under Grant 621QN270. *(Corresponding author: Qiang Fu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3367934

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self-supervised feature extractor to a Gaussian-like distribution to reduce the feature distribution mismatch problem. However, the low texture information, low resolution, and high noise level of thermal images limit the feature extraction capability of CNN [30]. The fusion of thermal and visible images with complementary information for object detection has achieved excellent performance in challenging scenarios. With the development of sensor technologies, the cost of obtaining thermal images has been reduced, and related research on cross-modal fusion based on thermal and visible for object detection has been promoted [31].

Most of the current work is designed for pedestrian detection tasks or general-purpose detection networks that cannot predict the orientation of targets in aerial imagery [32], [33], [34]. In addition, the studies are mostly conducted on natural scenes, which do not address the particular issues in aerial scenes, such as complex backgrounds, rotating and densely distributed targets, and extremely low visibility. As shown in Fig. 1, the horizontal boxes (red horizontal boxes) do not characterize the target orientation and carry a lot of redundant information, which is not friendly for densely distributed rotating targets. Imperfect manual labeling (within the green circle) causes targets with strong occlusion to be unlabeled resulting in the network not learning the occluded target features effectively. Some of the visible images in the Drone Vehicle dataset are almost completely black, making it impossible to distinguish the location of the vehicle. There are some rectangular unidentified objects (within the purple circles) in the thermal image that resemble the outline of the vehicle, which makes it difficult for the detection algorithm to distinguish them. However, multimodal object detection in UAV aerial images has just started [35], [36], [37], [38], [41], [42], and there are three main directions, in which methods are currently being investigated.

1) Fusion is performed in the backbone network part, and after fusion, only the fused modal features are learned, which lacks in-depth mining of the effective features of thermal and visible modalities.
2) An additional neural network fusion module is designed and embedded in the middle of the entire network to enhance the unique features of thermal and visible modes. The network adaptively selects the region of interest through the fusion module, and although differential features of both modalities are learned, some valuable shared features are eliminated and the sensitivity of visible to illumination cannot be circumvented.
3) Fusion methods use either summation or concatenation operations. The fused network uses the thermal modality as a baseline modality to guide the network's learning of the visible modality, an approach that ignores the advantages of the visible modality in daylight.

All these methods use thermal modality as a baseline modality, which makes it difficult to effectively address the existing problems mentioned above. In general, fully exploiting the advantageous features of each modality in different environments can effectively improve the object detection accuracy of aerial images.

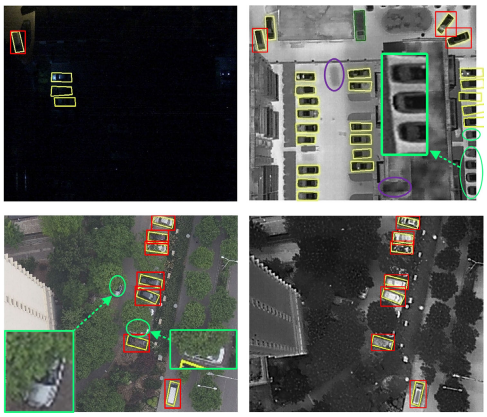


Fig. 1. Drone Vehicle dataset characteristics. The red box shows the effect of using the horizontal detection box. Compared with the rotated box (yellow box), the horizontal box has a lot of redundant information and is unable to characterize the object orientation. Green ellipses indicate the imperfect manual labeling, where objects with strong occlusion are not labeled, resulting in the network not being able to effectively learn such object features. Purple ellipses indicate rectangles similar to the objects.

In this study, our motivation is to propose a network that adaptively enhances the learning of dominant modality features that favor object detection for the current environment, fully exploiting the potential advantages of each modality to improve the accuracy of difficult object detection in aerial images. First, we propose an illumination perception module based on histogram statistics to better guide the object detection network to perform various feature processing tasks by judging the illumination conditions in the visible image scene and to mitigate the negative impact of visible images in low-light scenarios on the object detection network. Second, we design the uncertainty-aware module, which enables the thermal modality and the visible modality to play to their strengths and circumvent their negative impacts and quantifies various uncertainties existing in the two modalities into weights, so that the side of the two modalities that is favorable to target detection dominates in the object detection network. Then, in the design of the object detection network, to enhance the learning of difficult samples, we design a dual-stream backbone network based on the attention mechanism, and in the feature fusion part, we adopt the cross-modality attentive feature fusion (CMAFF) module to fully extract the shared and complementary features between the two modalities. To make full use of the advantages of the unique features of each modality, we especially design a three-branch feature enhancement network to enhance the learning of the visible modal features, the thermal modal features, and the fusion modal features, respectively. Finally, in the result fusion section, we design light perception nonmaximum suppression (LP-NMS) to reduce the interference of false-positive samples generated by the visible image detection branch and improve the horizontal detection head to a rotating one to preserve the orientation of the target and reduce redundant information.

The main contributions of this study are summarized as follows.

1) We propose a cross-modal oriented object detection network that combines a mid-term fusion method with a

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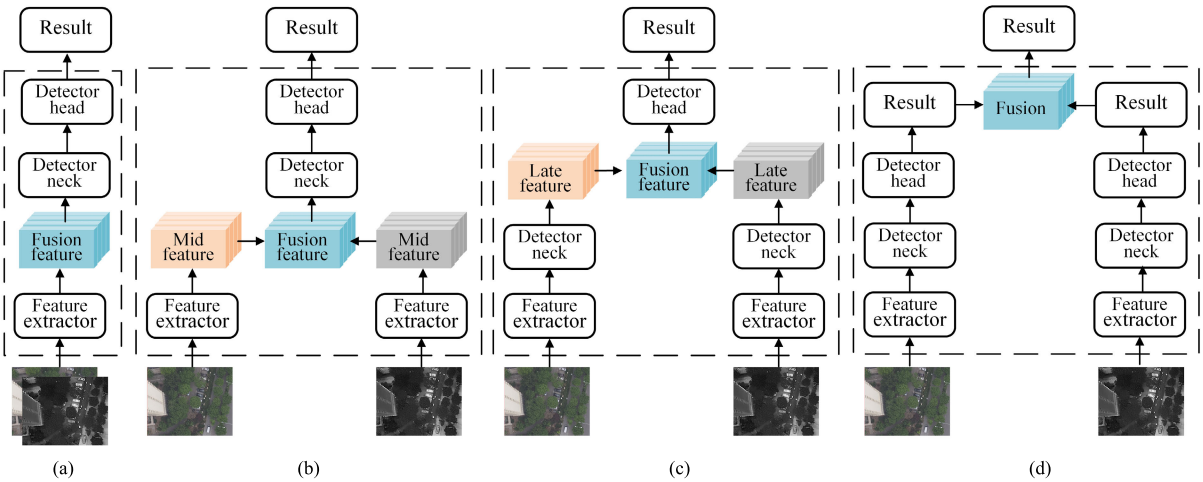


Fig. 2. Four fusion strategies. (a) Early fusion means that the fusion work is done at the input side, and fusion methods mainly use summation, concatenation, or pixelwise fusion. (b) Mid-term fusion refers to the fusion at a certain stage in the backbone, fusion methods besides using work such as summing and concatenation, some researchers design separate CNN fusion modules for feature fusion operations. (c) Late fusion refers to the fusion of features from two modalities at the end of the feature learning phase, and then feeding them together into the detection head. (d) Result fusion refers to the fusion of the results of two unimodal detectors (thermal modal detector and visible modal detector).

result fusion method, possessing high detection accuracy along with a lightweight network structure.

2) Our cross-modal aerial remote sensing image object detection (CRSIOD) network is the first cross-modal object detection network that no longer uses thermal modality as the reference modality, and it uses the illumination perception module and uncertainty-aware module to guide the network to fully exploit the advantageous features of each modality for different environments, which is beneficial for object detection.
3) We use YOLOv5s, a popular lightweight network in the industry, as the network framework and build a two-stream backbone network based on the attention mechanism, a three-branch feature enhancement network, and a rotating detection head. Our approach mines the effective information available for all three modalities in depth, effectively improving the network's ability to learn features for a limited number of samples per modality and improving detection performance while having a lightweight network architecture.
4) Extensive experiments on the three challenging publicly available datasets Drone Vehicle [38], a visible-infrared paired dataset for low-light vision (LLVIP) [39], and M^3FD [40] demonstrate that our method achieves state-of-the-art performance.

The rest of this article is organized as follows. In Section II, we will provide a brief introduction to the related work. Then, in Section III, the network structure and methodology will be described in detail. In Section IV, we will present our work, including experimental results and relevant comparisons, to validate the effectiveness of our approach. Finally, in Section V, a summary of the research findings will be provided.

II. Related Works

A. Advanced Detection Algorithms

To ensure the high performance of the algorithm, one usually trains the object detection network on a public, widely recognized dataset with high confidence, for instance, microsoft common objects in context (MS COCO) [43] and pascal visual object classes (PASCAL VOC) [44], training deep learning-based algorithms. The you only look once (YOLO) family of algorithms, with its simple network architecture, ultrafast running speed, and excellent performance, has been widely recognized and adopted in both academic and industrial fields. Until now, the YOLO family of algorithms is still being developed and applied in academia and industry [45], [46], [47], [48], [49], [50], [51]. Given the ease of development of YOLOv5 and its common use in engineering practice, we have chosen the small size of the YOLOv5 structure as our detection baseline.

The application of attention mechanisms in CNN can lead to a great improvement in network performance, and the main attention mechanisms commonly used are squeeze-and-excitation (SE) [52], bottleneck attention module (BAM) [53], convolutional block attention module (CBAM) [54], and coordinate attention (CA) [55]. SE ignores the location information of the target and only considers the channel information inside the network. BAM and CBAM introduce the location information by performing global pooling operation on the channel, but this method cannot obtain the global location information of the target, CA makes the network obtain large area information by embedding location information into the channel attention, and it is easier to capture the target features.

B. Multimodal Detection Approaches

In the early work, at which stage the features of the two modes are fused to give the best detector performance is the central question. Depending on the stage of fusion, cross-modal feature fusion methods can be divided into four forms: early fusion, mid-term fusion, late fusion, and result fusion [56], [57], as shown in Fig. 2. These two works show that it is better to employ mid-term fusion when performing feature fusion for both one-stage scheme and two-stage scheme. Fusion methods often use direct summation or concatenation [35], [38], which lack the combing of features and the mining of important features.

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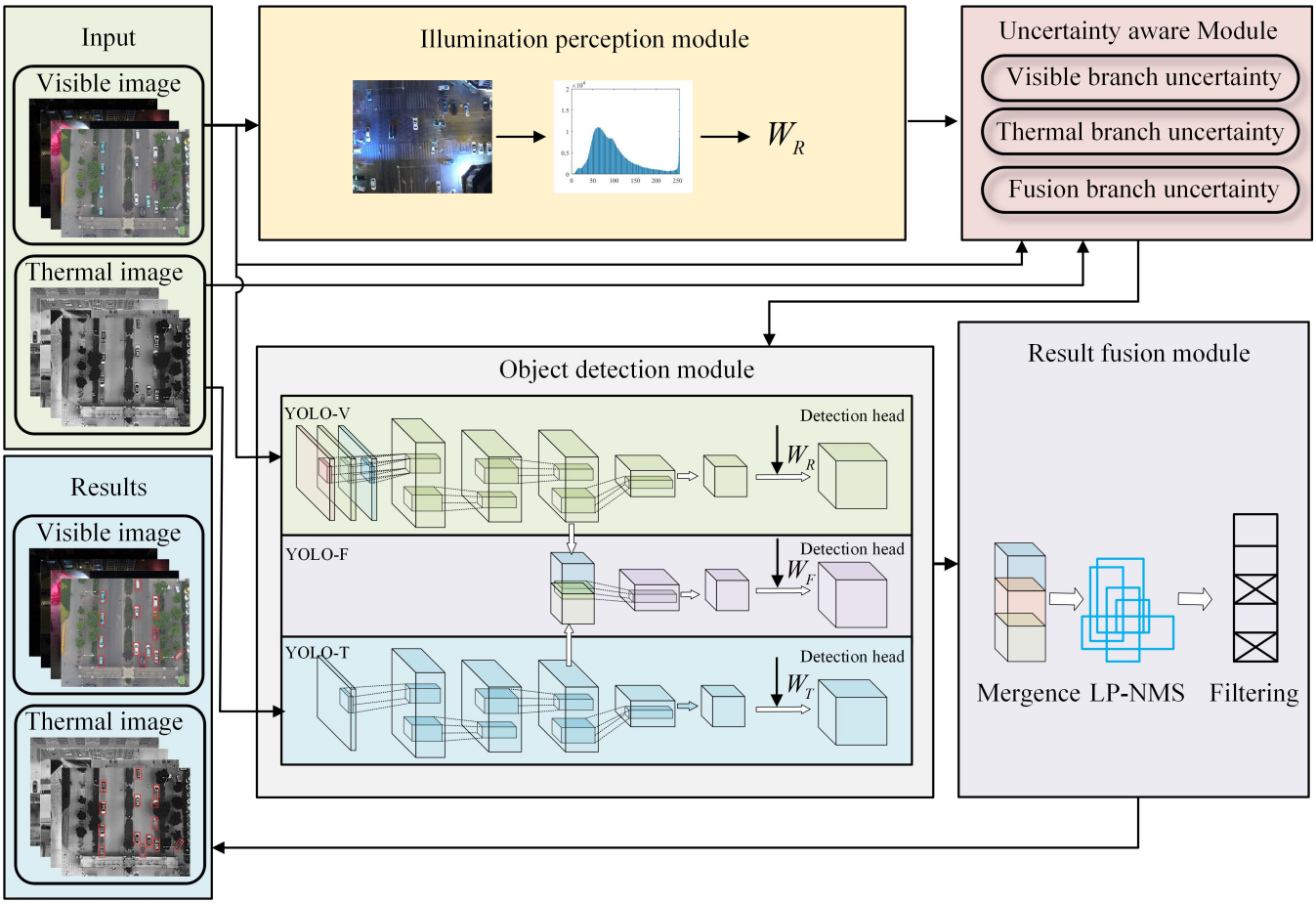


Fig. 3. Overall framework of CRSIOD. The CRSIOD consists of four main components. The first part is the illumination perception module, which is used to distinguish different lighting environments in visible images and guide the detection network to perform various feature processing tasks. The second part is the uncertainty-aware module, which is used to quantify various uncertainties of each target to facilitate the optimization of the detection model. The third part is the object detection module, YOLO-V is responsible for learning visible modal features, YOLO-T is responsible for learning thermal modal features, and YOLO-F is responsible for learning fused modal features. The fourth part is the result fusion module for optimizing the detection results of the object detection module.

Some related works have pointed out the fact that good fusion methods can effectively improve object detection performance [36], [37], [41], [42]. Fang and Wang [36] proposed a CMAFF fusion module inspired by differential amplifiers, which showed good performance, but it neglected to make full use of the thermal and visible modal features and poor detection of nighttime effects. Wang et al. [37] proposed an illumination perception module to cope with nighttime conditions, but the use of a neural network design increases the complexity of the network. Zhang et al. [41] and Zhu et al. [42] mainly focus on the design of network lightweight, and the detection object is a very small target in high-altitude scenes, which cannot cope with the densely distributed targets with different scales in complex scenes in medium and high altitudes. It is worth noting that these methods either focus on the design of the fusion module or on the design of the lightweight. In short, to our knowledge, none of the existing methods is designed for the characteristics of the image itself.

III. Proposed Method

A. Overall CRSIOD Architecture

The overall architecture of the proposed CRSIOD is shown in Fig. 3. The CRSIOD consists of four main components: illumination perception module, uncertainty-aware module, object detection module, and result fusion module. The illumination perception module is used to distinguish different illumination environments in visible images and guide the detection network in various feature processing tasks. The uncertainty-aware module converts various uncertainties of the two modality objectives into weights to facilitate the optimization of the detection model. The object detection module detects targets in the image by combining illumination information from the illumination perception module and uncertainty weights from the uncertainty aware module for each object. The result fusion module combines and optimizes the detection results from the object detection module and finally outputs the final detection result.

B. Illumination Perception Module

Photographers will make good use of the histogram in the camera when shooting, as it can reflect the exposure level and scene characteristics of that image. Motivated by this, we use histograms of visible images as a basis to design the illumination perception module.

As shown in Fig. 4, the illumination is sufficient during the day, and the pixels are mainly distributed between 50 and 200,

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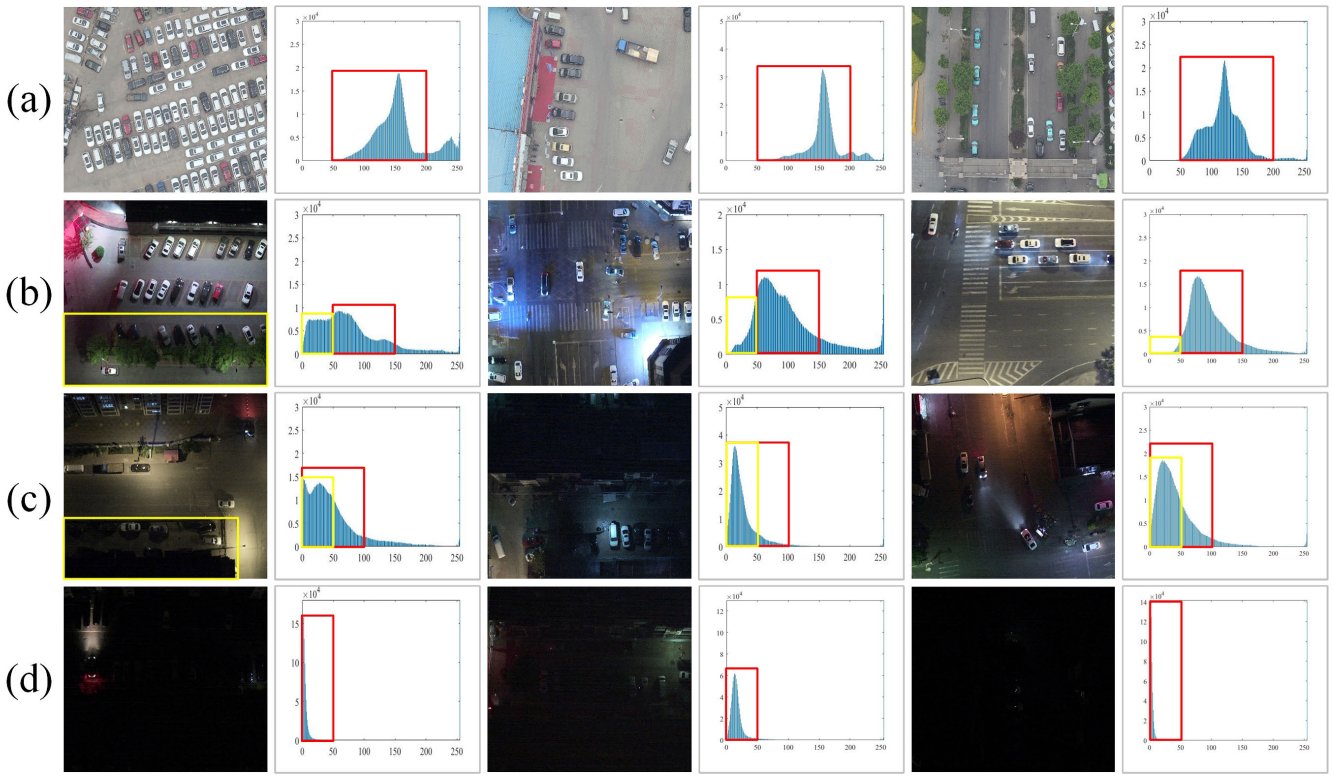


Fig. 4. Visible images and their corresponding histograms for different illumination and scenarios. (a) During the daytime, when there is sufficient illumination, the pixels are mainly distributed between 50 and 200 (in the red boxes). (b) Night bright illumination, more adequate illumination, pixels are mainly distributed between 50 and 150 (red boxes). When the distribution of pixels between 0 and 50 accounted for half of all pixels distributed (yellow boxes) indicates that the illumination is insufficient or the scene is more complex. (c) Low illumination at night, insufficient illumination, and complex scenes, pixels are mainly distributed between 0 and 100 (in red boxes), similar to night bright illumination, when the distribution of pixels between 0 and 50 (in yellow boxes) accounts for an increasing proportion of all the pixels distributed indicates that the illumination intensity is getting weaker and weaker. (d) Dark night, weak or no illumination, no scene information, pixels are mainly distributed between 0 and 50 (in the red boxes).

when the image scene information are rich, the target color and texture details are clear, and the feature is obvious, which can compensate for the shortcomings of the lack of thermal image scene information, so the illumination weight value is set to 1. Strong nighttime illumination provides the adequate illumination with most pixels distributed between 50 and 150. There are also a few slightly fainter pixels distributed between 0 and 50, but the ratio is small. By observing the image, we can see that when the pixels between 0 and 50 account for half of all the pixels, this indicates that the illumination may be insufficient or the scene is complex. The target may be submerged in a slightly darker environment or obscured. In this case, assistance from the thermal detection branch is required. Therefore, setting the illumination weight is divided into two cases: 1) $1-(v/s)$ when the proportion of pixels (v) between 0 and 50 to all pixels (s) is more than an illumination perception threshold (illu_thre) and 2) otherwise, it is set to 1. Low illumination at night, insufficient illumination, complex scenes, pixels mainly distributed between 0 and 100, a small part of the target texture information color information retention, target features are not obvious, so set the illumination weight value: $1-(v/s)$. Black night, weak or no illumination, no scene information, pixels are mainly distributed between 0 and 50, target features are basically lost, at this time mainly rely on thermal image detection branch, so set the illumination weight value as $1-(v/s)$. In summary, the illumination perception module W_R can be represented as follows:

$$W_R = \begin{cases} 1-\dfrac{v}{s}, & \dfrac{v}{s} \geq \text{illu_thre} \\ 1, & \dfrac{v}{s} < \text{illu_thre}. \end{cases} \tag{1}$$

The illumination threshold (illu_thre) was determined by a genetic algorithm [58] with an optimal value of 0.50.

C. Uncertainty-Aware Module

As shown in Fig. 1, in aerial image scenarios, due to the complex environment, there are many uncertainties in the detection process of targets [38], which are summarized as follows.

1) There are some rectangular unidentified objects in the thermal image that resemble the outline of the vehicle, which makes it difficult for the detection algorithm to distinguish them.
2) Due to the imperfection of the thermal imaging equipment, it is also easy to appear in the thermal image screen similar to the shadow of the vehicle.
3) The thermal and visible image pairs are obtained by postprocessing, and it is difficult to achieve complete pixel-level alignment.
4) Because objects in visible images are often missed annotations in low-light conditions, it is difficult to match the exact location between visible and thermal images.

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TABLE I

QUANTITATIVE TABLE OF UNCERTAINTIES PRESENT IN THE DRONE VEHICLE DATASET: 1) RECTANGULAR UNIDENTIFIED OBJECT IN THE THERMAL IMAGE; 2) VEHICLE-LIKE SHADOW IN THE THERMAL IMAGE DUE TO IMPERFECT THERMAL IMAGING DEVICES; 3)THERMAL AND VISIBLE IMAGE PAIRS ARE NOT MATCHED TO THE TARGET BOUNDING BOX; AND 4) MISSING LABELED OBJECT IN THE VISIBLE IMAGE

Uncertainty factor	Number	percentage
1)	89950	18%
2)	143920	29%
3)	190195	38%
4)	47947	10%

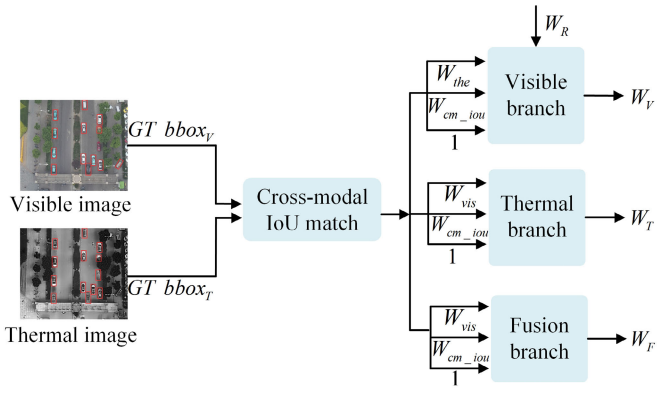


Fig. 5. Uncertainty-aware module. First, the CM_IoU of the real object boxes in the thermal and visible image pairs is calculated as the uncertainty of whether the object positions of the two modalities are matched or not, and then, the uncertainty weights for the matching of the object positions are inputted into the branches of the three modalities (thermal, visible, and fusion), and since the visible image is susceptible to the influence of the illumination factors, the uncertainty weights of the illumination situation (W_R) are added to the visible detection branch, and finally, the uncertainty factor weights of the visible modal uncertainty (W_V), the thermal modal uncertainty (W_T), and the uncertainty factor weights of the feature fusion branch (W_F) are generated, respectively.

We quantify these uncertainties in Table I.

The uncertainty-aware module is used to convert various uncertainties of the two modality objectives into weights, and the specific design is shown in Fig. 5.

The uncertainty-aware module first computes the cross-modal intersection over union (CM_IoU) of the real object boxes in the visible–thermal image pairs separately. It can determine the difference in position between the two modal counterpart targets. CM_IoU can be calculated as follows:

$$\text{CM_IoU} = \frac{f_{\text{area}}(B_V \cap B_T)}{f_{\text{area}}(B_V \cup B_T)} \tag{2}$$

where B_V and B_T denote the real target boxes in the visible and thermal images, respectively. The subscript area indicates the area of the polygon.

In daytime scenes with sufficient illumination, for thermal images, it is easy to miss the object during manual annotation due to the lack of color information and texture information. To overcome this problem, we use the real object boxes in the visible modality to fill these missing object boxes in the thermal modality. Then, we set an uncertainty weight W_{vis} for the newly added object boxes in the thermal modality, while the weight of the original object boxes in the thermal modality is set to 1.

Moreover, it is difficult to ensure that the cross-modal image pairs are fully aligned pixel-by-pixel during image acquisition. During the daytime, when there is sufficient illumination, the visible image can clearly characterize the object features, which is more advantageous for object detection; therefore, we use the object position in the visible image as a reference and set an alignment threshold μ. When $0 < \text{CM_IoU} < \mu$, we use CM_IoU as the misaligned object's uncertainty weight $W_{\text{cm_iou}}$ and assign it to the true object box of the object in the thermal modality in the daytime scene. Finally, the uncertainty weight W_T for each object in the thermal modality in the daytime scenes is calculated as shown in the following equation:

$$W_T = \begin{cases} W_{\text{vis}}, & \text{if thermal object miss} \\ 1 \times W_{\text{cm_iou}}, & \text{otherwise.} \end{cases} \tag{3}$$

In the night both night bright light (in the night bright light scenes although the object still has obvious color and texture information in the visible image, there are still dark areas, this part of the dark area may lead to the object missing detection) and low light at night need to be assisted by the thermal image detection branch, especially in the dark night scenes, the thermal image detection branch will be the dominant role. Therefore, the uncertainty weight W_T for each object in the thermal modal is calculated as shown in the following equation in the night scenes:

$$W_T = \begin{cases} W_{\text{vis}}, & \text{if thermal object miss} \\ 1, & \text{otherwise.} \end{cases} \tag{4}$$

For visible images, there is enough illumination during the daytime, and visible images can clearly characterize the object features, which is more advantageous for object detection, but there may also be omissions in manual annotation, so we use the real object boxes in the thermal modality to fill these missing object boxes in the visible modality. We set an uncertainty weight W_{the} for the newly added object boxes in the visible modality, while the weight of the original object boxes in the visible modality was set to 1. Therefore, in the daytime scenario, the uncertainty weight W_V for each object in the visible modality is calculated as shown in the following equation:

$$W_V = \begin{cases} W_{\text{the}}, & \text{if visible object miss} \\ 1, & \text{otherwise.} \end{cases} \tag{5}$$

In the night scenes, due to the illumination sensitivity of the visible image, it is necessary to use the assistance of the thermal image detection branch. The strength of the auxiliary role with the gradual weakening of the illumination intensity and enhancement, in the dark of the night illumination weak or no illumination scenes, the thermal image detection branch will play a dominant role in the illumination intensity. We use the illumination perception module output of the illumination weight value W_R to characterize the illumination intensity. In night scenes, the detection branch of the thermal image is more dominant, and the object position in the thermal

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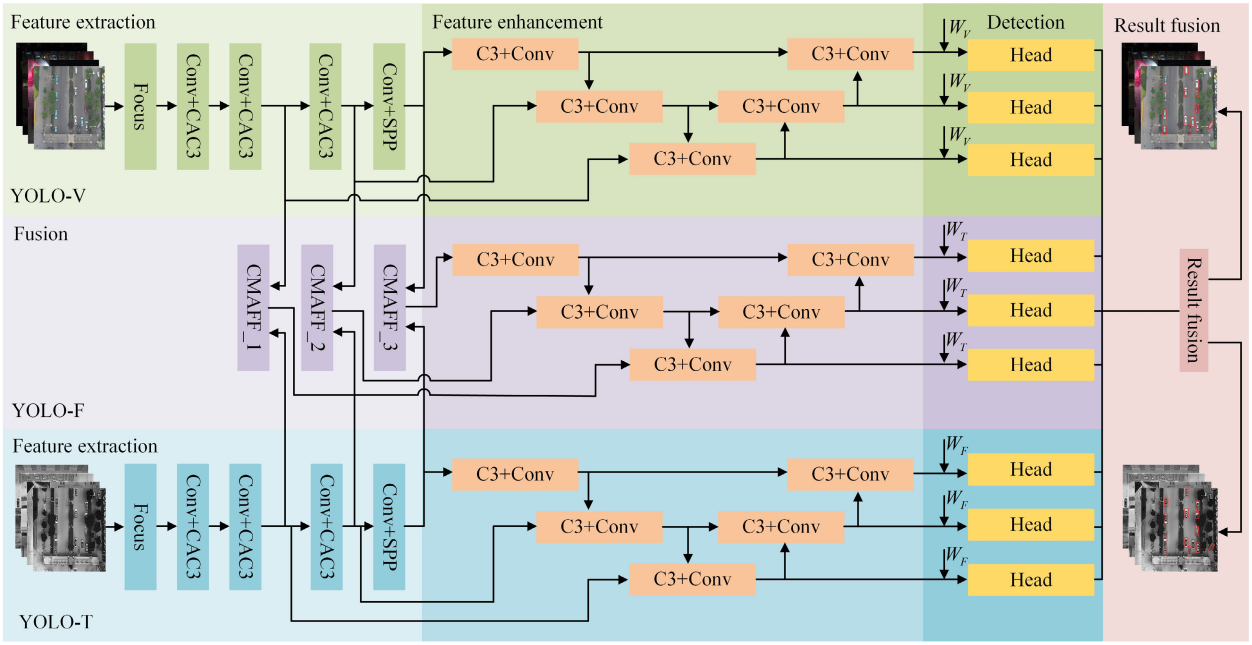


Fig. 6. Object detector module structure diagram. The feature extraction part is responsible for extracting and learning the features of thermal and visible modalities, while the fusion part gradually fuses the features of thermal and visible modalities from shallow to deep to fully exploit the shallow contour features and deep semantic features of both modalities. The feature enhancement part augments and learns thermal, visible, and fused modal features, respectively. The feature pyramid network (FPN) + path aggregation network (PAN) structure is adopted, where the FPN layers transmit strong semantic features from top to bottom, while the PAN layers transmit strong localization features from bottom to top, and the learning of objects at different scales is achieved by the aggregation of features from different layers. The head part is responsible for predicting object coordinates, categories, and confidence level information. The result fusion part is responsible for merging the results output from the detection head and optimizing them. The green area represents YOLO-V (visible mode detection branch), the purple area represents YOLO-F (fusion mode detection branch), and the blue area represents YOLO-T (thermal mode detection branch).

image is more prominent with respect to the visible image during manual annotation. Therefore, we use the position of the object in the thermal image as a reference to determine the uncertainty weight $W_{\text{cm_iou}}$ of the misaligned object and assign it to the true object box of the object in the visible modality of the night scenes. Therefore, in the night scenario, the uncertainty weight W_V for each object in the visible modality is calculated as shown in the following equation:

$$W_V = \begin{cases} W_{\text{the}}, & \text{if visible object miss} \\ W_{\text{cm_iou}} \times W_R, & \text{if not alignment} \\ W_R, & \text{if alignment.} \end{cases} \tag{6}$$

In the daytime scenes, visible images can clearly characterize the object features, which is more advantageous for object detection and more accurate for manual annotation. In night scenes, the object features in thermal images are more prominent, which is more favorable for object detection and more accurate for manual annotation. Thus, the uncertainty weight W_F for each object box on the fusion branch is consistent with that calculated for visible modality in the daytime scenes and thermal modality in the nighttime scenes as shown in the following equation:

$$W_F = \begin{cases} W_V, & \text{in the daytime scenes} \\ W_T, & \text{in the night scenes.} \end{cases} \tag{7}$$

D. Object Detection Module

In the previous literature, the unique features and target characteristics of each sensor are not fully utilized, leading to the performance degradation. Therefore, for the detection of difficult samples in complex scenes, such as strong occlusion and low light, we design a dual-stream backbone network based on the CA mechanism to enhance the network's focus on difficult samples and suppress the interference of complex backgrounds and input the three modalities into the feature enhancement network separately after extracting the fused modalities using the CMAFF module to make full use of the advantages of the unique features of each modality. In the design of the detection head, we improve the horizontal detection bounding box to a rotating detection bounding box to preserve the object orientation. The overall structure of the object detection module is shown in Fig. 6.

As shown in Fig. 7, in order to avoid the loss of location information caused by the global pooling operation, CA focuses on two dimensions, width and height, respectively, to facilitate the effective use of the spatial coordinate information of the input feature map. CA contains two parts: information embedding and attention generation. In the information embedding stage, the global average pooling operation is performed for all channels of the input feature map along the vertical coordinate direction and horizontal coordinate direction, respectively, to obtain $c \times h \times 1$ and $c \times w \times 1$ feature maps, and then, the feature convergence is input to the attention generation stage. In the attention generation stage, the input feature map is first stitched into a $c \times 1 \times (h + w)$ feature map, and the feature map is compressed from c dimension to c/r dimension by shrinking rate r. Second, the feature map is decomposed into a $c/r \times 1 \times w$ horizontal attention tensor and a $c/r \times 1 \times h$ vertical attention

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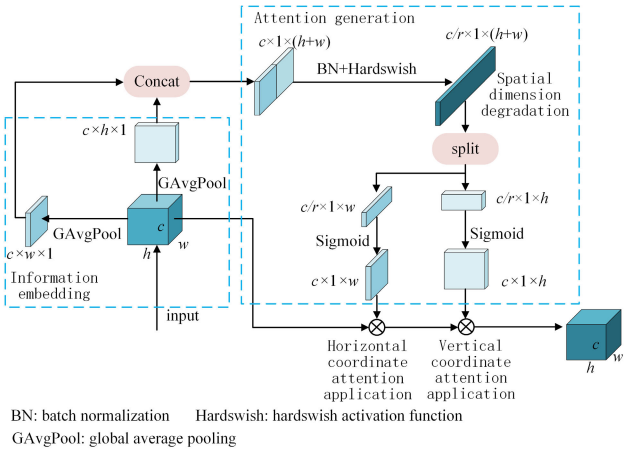


Fig. 7. CA mechanism. CA consists of two parts: information embedding and attention generation. The information embedding splits the feature map into width and height dimensions, which facilitates effective utilization of the spatial coordinate information of the input feature map. The attention generation part multiplies the features in width and height dimensions with the original input feature map to achieve strong attention to object location.

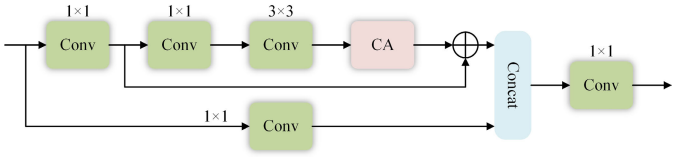


Fig. 8. CAC3 module structure diagram.

tensor along the spatial dimension, and then, it is expanded into $c \times 1 \times w$ and $c \times 1 \times h$, respectively, and finally multiplied with the input feature map.

To enhance the detection of more difficult samples, such as strong occlusion, dense distribution, and obscure features in visible and thermal images, we added the CAC3 module in both features extraction networks to assign higher weights to the more difficult vehicle targets to increase the attention to them and assign lower weights to suppress the natural background, as shown in Fig. 8.

Cross-modal fusion typically involves the extraction of shared and unique features between two modalities. The traditional fusion method using direct summation or concatenation has been proven ineffective in extracting such features. In order to address these limitations, a cross-modal fusion module (CMAFF) was proposed in the literature [36]. CMAFF has been shown to achieve better performance in the multimodal features fusion of remote sensing images. Therefore, in this article, we utilize the CMAFF module in the multimodal features fusion method. The specific design is shown in Fig. 9. The unique features enhancement module and the shared features selection module make up the cross-modal feature fusion module.

The computation of the CMAFF module proceeds as follows:

$$\begin{aligned} F^V &= \frac{F^V + F^V}{2} + \frac{F^T - F^T}{2} \\ &= \frac{F^V + F^T}{2} + \frac{F^V - F^T}{2} \\ &= \frac{F^S}{2} + \frac{F^U}{2} \end{aligned} \tag{8}$$

$$\begin{aligned} F^T &= \frac{F^T + F^T}{2} + \frac{F^V - F^V}{2} \\ &= \frac{F^V + F^T}{2} - \frac{F^V - F^T}{2} \\ &= \frac{F^S}{2} - \frac{F^U}{2} \end{aligned} \tag{9}$$

$$F^S = F^V + F^T \tag{10}$$

$$F^U = F^V - F^T \tag{11}$$

where F^V is for the visible convolutional maps, F^T is for the thermal convolutional maps, F^S denotes the shared features, and F^U is the unique features of the two modes.

As shown in Fig. 10, for the unique features module, first, the visible convolutional feature maps $F^V \in \mathbb{R}^{c\times h\times w}$ and the thermal convolutional feature maps $F^T \in \mathbb{R}^{c\times h\times w}$ are directly subtracted to obtain the unique feature maps $F^U \in \mathbb{R}^{c\times h\times w}$. Second, the attention maps of channel dimension $M_{\mathrm{UE}} \in \mathbb{R}^{c\times 1\times 1}$ are inferred from the unique feature maps $F^U \in \mathbb{R}^{c\times h\times w}$, to achieve feature adaptive refinement, the input feature maps are multiplied with the refined feature maps. Third, the enhanced feature maps are created by combining the original input feature maps for each mode with the refined feature maps. Finally, to create the unique enhanced feature map $F_{\mathrm{UE}} \in \mathbb{R}^{c\times h\times w}$, two improved feature maps are added together. The following is how this process is calculated:

$$F_{\mathrm{UE}} = F^V \otimes (1 + M_{\mathrm{UE}}) + F^T \otimes (1 + M_{\mathrm{UE}}) \tag{12}$$

where

$$S_1 = f_{\mathrm{GAP}}(F^U) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} F^U(i, j) \tag{13}$$

$$S_2 = f_{\mathrm{GMP}}(F^U) = \max(F^U(i, j)) \tag{14}$$

$$Z_1 = \mathrm{Conv}_2(R(\mathrm{Conv}_1(S_1))) \tag{15}$$

$$Z_2 = \mathrm{Conv}_2(R(\mathrm{Conv}_1(S_2))) \tag{16}$$

$$M_{\mathrm{UE}} = \sigma(Z_1 + Z_2). \tag{17}$$

The superscripts V and T denote the visible and thermal modalities, respectively; subscript UE indicates that the variable belongs to the unique enhancement module; $\otimes$ denotes the elementwise multiplication; σ indicates the sigmoid function; $R(.)$ denotes the rectified linear unit (ReLU) activation function; $Z_1 \in \mathbb{R}^{c\times 1\times 1}$, $Z_2 \in \mathbb{R}^{c\times 1\times 1}$, $S_1 \in \mathbb{R}^{c\times 1\times 1}$, and $S_2 \in \mathbb{R}^{c\times 1\times 1}$ are the four intermediate variables; Conv_1 and Conv_2 mean the embedded function by 1×1 convolution; f_{GAP} is considered as the global average pooling; superscript U represents the differential operation; and f_{GMP} is considered as the global max pooling.

The shared features selective module adaptively selects the channel features of both modalities based on the shared modality features and mixes them into a new feature, as shown in Fig. 11.

As shown in Fig. 11, for the shared features module, first, the visible convolutional feature maps $F^V \in \mathbb{R}^{c\times h\times w}$ and the thermal convolutional feature maps $F^T \in \mathbb{R}^{c\times h\times w}$ are directly summed to get the co-modal features $F^S \in \mathbb{R}^{c\times h\times w}$; second, the co-modal feature maps are passed through a global average

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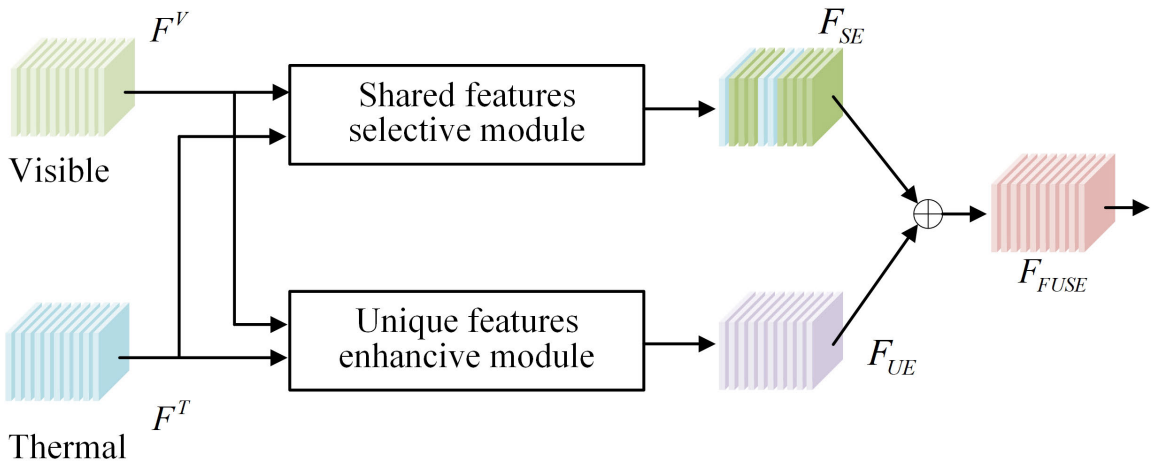


Fig. 9. Overall structure of the cross-modal fusion module (CMAFF). The shared feature selection module and the unique feature enhancement module are implemented in parallel.

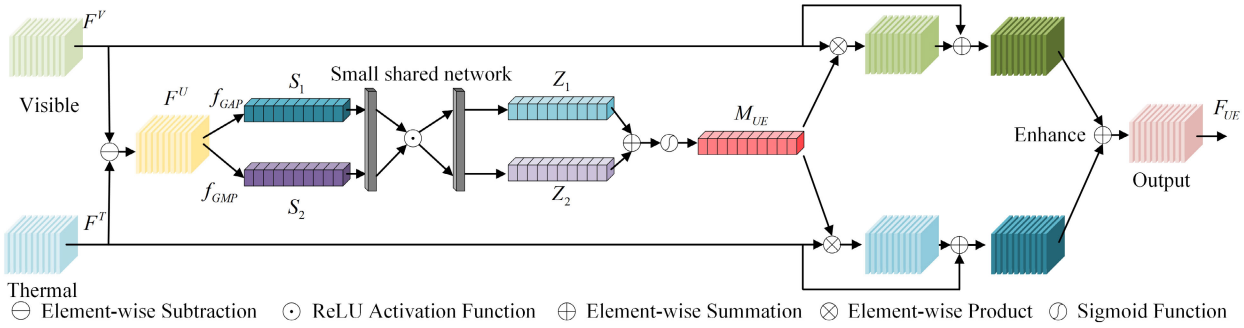


Fig. 10. Overall structure of the unique feature enhancement module. The feature attention map M_{UE} is obtained from the differential features of the two modalities, the feature refinement is performed by multiplying M_{UE} with the original modality features, and the refined feature map is summed with the original modalities for feature enhancement.

pooling layer f_{GAP} and two fully connected layers f_{FC}, and filtered by the softmax function to obtain the visible feature attention maps $M_{\mathrm{SE}}^V \in \mathbb{R}^{c\times 1\times 1}$ and thermal feature attention maps $M_{\mathrm{SE}}^T \in \mathbb{R}^{c\times 1\times 1}$; third, multiplying the visible and thermal feature attention maps with their inputs (F^V, F^T), respectively, to obtain the refined feature maps; and finally, the two refined feature maps are summed to get the output $F_{\mathrm{SE}} \in \mathbb{R}^{c\times h\times w}$ of the shared selection module. This procedure can be formulated as follows:

$$F_{\mathrm{SE}} = F^V \otimes M_{\mathrm{SE}}^V + F^T \otimes M_{\mathrm{SE}}^T \tag{18}$$

where

$$S = f_{\mathrm{GAP}}(F^S) = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W} F^S(i,j) \tag{19}$$

$$Z_1 = f_{FC_1}(S) \tag{20}$$

$$Z_2 = f_{FC_2}(S) \tag{21}$$

$$M_{\mathrm{SE}}^V = f_{\mathrm{soft\,max}}(Z_1) \tag{22}$$

$$M_{\mathrm{SE}}^T = f_{\mathrm{soft\,max}}(Z_2). \tag{23}$$

The subscript SE indicates that the variable is subordinate to the shared features selection module. The superscript S represents the shared modality. $Z_1 \in \mathbb{R}^{c\times 1\times 1}$, $Z_2 \in \mathbb{R}^{c\times 1\times 1}$, and $S \in \mathbb{R}^{c\times 1\times 1}$ denote the three intermediate features. f_{FC_1} and f_{FC_2} denote the two fully connected networks with two layers sharing the first layer weights. f_{softmax} refers to softmax operation.

The combination of the unique features enhancement module and the shared features selection module is the last step to the fusion of the visible and thermal features. The final calculation process is shown as follows:

$$\begin{aligned} F_{\mathrm{FUSE}} &= F_{\mathrm{UE}} + F_{\mathrm{SE}} \\ &= F^V \otimes (1 + M_{\mathrm{UE}}) + F^T \otimes (1 + M_{\mathrm{UE}}) \\ &\quad + F^V \otimes M_{\mathrm{SE}}^V + F^T \otimes M_{\mathrm{SE}}^T \\ &= F^V \otimes (1 + M_{\mathrm{UE}} + M_{\mathrm{SE}}^V) \\ &\quad + F^T \otimes \left(1 + M_{\mathrm{UE}} + M_{\mathrm{SE}}^T\right). \end{aligned} \tag{24}$$

The loss function of YOLOv5 employs a multitask loss that uses a horizontal bounding box to frame the detected target, but the horizontal bounding box does not characterize the orientation of the target and carries a lot of redundant information, which is not suitable for densely distributed rotating targets. Therefore, in order to characterize the orientation of the target, we add an angle classification layer and an angle classification loss to the CRSIOD head structure in order to adapt to the characteristics of the target of the aerial scene image. The loss function of CRSIOD consists of four components, in terms of confidence loss, classification loss, angular classification loss, and target bounding box regression loss.

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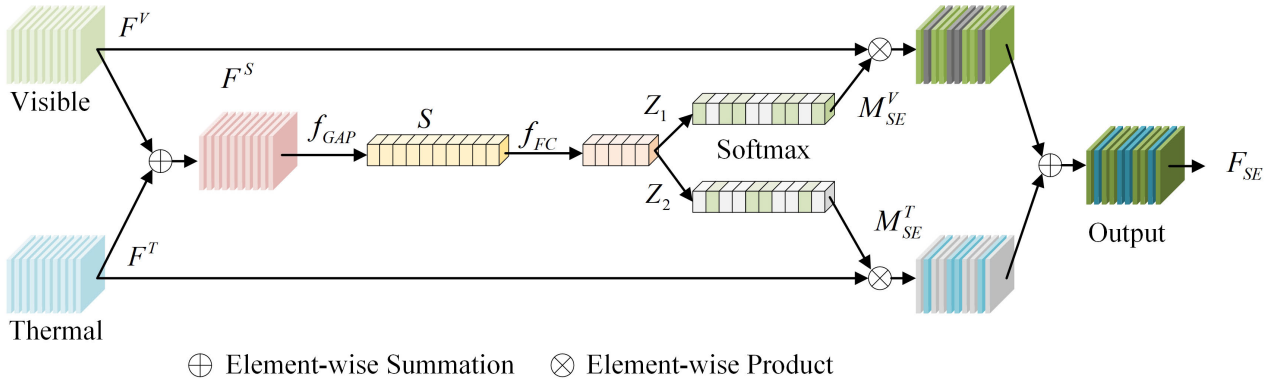


Fig. 11. First, the thermal image is summed with the visible image to get the co-modal feature maps, the co-modal feature maps are filtered by Softmax to get the feature attention maps of two modes, which are multiplied with the original feature maps to get the refined feature maps of each mode, and finally, the refined feature maps of the two modes are summed up to get the output of the final shared feature selection module.

The BCEWithLogitsLoss loss function [59] is used for both confidence loss l_{obj} and classification loss l_{cls}, defined as follows:

$$l_{n,c} = -\frac{1}{n}\sum\left[\begin{array}{c} y_{n,c}\cdot\log\sigma(x_{n,c})+ \\ (1-y_{n,c})\cdot\log(1-\sigma(x_{n,c}))\end{array}\right] \tag{25}$$

where n is the number of the samples in the batch; $x_{n,c}$ represents the predicted result, $y_{n,c}$ ground-truth result; and σ is a sigmoid function.

Since the angle is treated as a classification task, the target box regression loss l_{bbox} is still calculated using the horizontal bounding box CIoU loss [60], which can be expressed as follows:

$$L_{\text{CIoU}} = 1 - \text{IoU} + \frac{\rho^2(b, b^{gt})}{d^2} + \alpha v \tag{26}$$

where

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w^{gt}}{h^{gt}} - \arctan\frac{w}{h}\right)^2 \tag{27}$$

$$\alpha = \frac{v}{(1-\text{IoU})+v} \tag{28}$$

where $\rho^2(b, b^{gt})$ denotes the Euclidean distance between the center point of the prediction result and the center point of the ground-truth bounding box; d represents the diagonal distance of the smallest closed region that can contain both the predicted and the true box; α is the weighting factor; and v is used to measure the consistency of the relative proportions of the width w and height h of two rectangular boxes.

In order to obtain precise orientation and scale information of the detected targets, we use a five-parameter long side labeling form (x_c, y_c, l, s, θ) to label the objects. (x_c, y_c) denotes the coordinates of the center point of the target bounding box, l represents the long side of the target bounding box, s denotes the short side of the target bounding box, and θ denotes the acute angle between the long side l and the x-axis of the coordinate axis. We refer to the literature [61], in order to avoid the boundary problem caused by the periodicity of the angle (which can lead to a sudden increase in the loss value at the boundary of the periodic change of the angle and increase the learning difficulty of the network), we use the circular smoothing labeling (CSL) + long edge definition method to transform the angular regression problem into an angular classification problem. The categorization details are shown in Fig. 12. We categorize the angle θ into 180 classes, and the specific expression for CSL is shown in the following equation:

$$\text{CSL}(x) = \begin{cases} g(x), & \theta - r < x < \theta + r \\ 0, & \text{otherwise.} \end{cases} \tag{29}$$

Algorithm 1 Pseudocode of the Fusion Filter

Input: $B^1, S^1, B^2, S^2, B^3, S^3$
B^1, B^2, B^3 are the list of detection boxes of the visible image object detector, the thermal image object detector and the modal features fusion object detector
S^1, S^2, S^3 are corresponding confidence scores of the visible image object detector, the thermal image object detector and the modal features fusion object detector
N_t is the threshold of NMS
W_R is the illumination uncertainty weight
Output: D, S

```
// 1. mergence
D ← {}
S^1 ← S^1 × W_R
B ← B^1 ∪ B^2 ∪ B^3
S ← S^1 ∪ S^2 ∪ S^3
// 2. LP-NMS
while B ≠ ∅ do
    m ← arg max S
    M ← b_m
    D ← D ∪ M; B ← B − M
    for b_i in B do
        if IoU(M, b_i) ≥ N_t then
            B ← B − b_i; S ← S − s_i
        end if
    end for
end while
// 3. filtering
for (b_i, s_i) in (D, S) do
    if s_i < 0.5 then
        D ← D − b_i; S ← S − s_i
    end if
end for
return D, S
```

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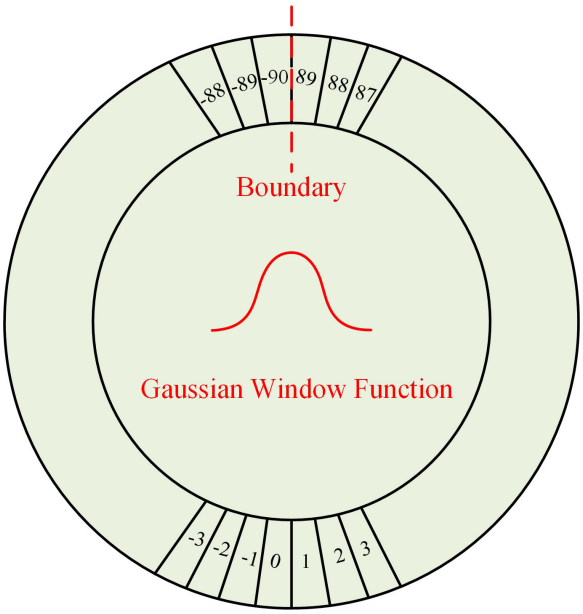


Fig. 12. Detailed diagram of the CSL implementation. We divided the angles into 180 classes, and the window function was chosen to be a Gaussian.

where

$$g(x) = \exp\left(-\frac{(x-\theta)^2}{2r^2}\right) \tag{30}$$

where $g(x)$ is the window function. Reference to the literature [61], we use a Gaussian window function with a window function radius r of 1. θ denotes the rotation angle of the current bounding box. The angles are processed by CSL and the angular loss l_θ is calculated using the BCEWithLogitsLoss loss function.

Therefore, the total loss function of a single branch can be expressed as follows:

$$L_i = w_i(l_{\text{obj}} + l_{\text{cls}} + l_{\text{bbox}} + l_\theta) \tag{31}$$

where $i \in \{\text{visible}, \text{thermal}, \text{fusion}\}$, w_i represents the uncertainty weight.

The total loss function is

$$L = L_{\text{visible}} + L_{\text{thermal}} + L_{\text{fusion}}. \tag{32}$$

E. Result Fusion Module

The result fusion module consists of three steps: merging, the light perception LP-NMS, and filtering. In the merging step, the results from the three branches are combined. Then, the LP-NMS is applied to eliminate redundant information. Finally, the screening process is carried out to refine the merged results.

In the mergence step, the detection results of the visible image object detector, the thermal image object detector, and the modal features fusion object detector are merged directly to obtain the preliminary fusion results.

Considering the sensitivity of the visible images to illumination and the effectiveness of NMS [62] in reducing redundant bounding boxes, light perception LP-NMS is proposed. In LP-NMS, we have assigned the preliminary prediction boxes and the corresponding detection scores for each of the three detection branches: $B^1 = \{b_1^1, \ldots, b_n^1\}$, $B^2 = \{b_1^2, \ldots, b_n^2\}$, $B^3 = \{b_1^3, \ldots, b_n^3\}$, $S^1 = \{s_1^1, \ldots, s_n^1\}$, $S^2 = \{s_1^2, \ldots, s_n^2\}$, and $S^3 = \{s_1^3, \ldots, s_n^3\}$. We multiply the detection score of the predicted target box of the visible modality by the current illumination condition W_R in the scene. Then, the redundant detection results can be removed by comparing the IoU of the actual target box with the predicted target box using a predefined threshold N_t ($N_t \in \mathbb{R}^{1\times1}$).

In the final filtering step, the fusion module generates the final results of fusion by excluding the results of detection with confidence scores lower than 0.5. The pseudocode for the resulting fusion module is presented in Algorithm 1.

IV. Experiment and Analysis

In this section, we conduct experiments on the Drone Vehicle dataset [38] to evaluate the performance of the CRSIOD algorithm. We detail our experimental setup, ablation study, and comparison with competing algorithms.

A. Dataset Introduction

The Drone Vehicle dataset contains 28 439 visible–thermal image pairs of vehicles captured by camera-equipped UAV. Several samples from the dataset are depicted in Fig. 13. These images were taken on urban roads, residential areas, and highways, with lighting conditions ranging from dark to night to day. The shooting heights were 80, 100, and 120 m, and the shooting angles were 15°, 35°, and 45°. During the dataset calibration process, cross-modal image pairs are aligned through affine and regional cropping techniques.

As shown in Fig. 14, there is an imbalance in the categories of the Drone Vehicle dataset, with the majority of targets being small. This poses a greater challenge for detection. To address this issue, we improve the mosaic data augmentation method for the light-aware mosaic data augmentation method. This method can boost the number of small samples and increase the complexity of the scene.

B. Implementation Details

All experiments in this article were implemented in Pytorch [63] on NVIDIA GeForce RTX 2080Ti GPU, and its operating system is ubuntu18.04, and the CUDA version 10.0. We choose a stochastic gradient descent (SGD) optimizer with a momentum of 0.843 in our experiments, and the learning rate decreases in a cos manner, varying periodically from 0.0032 to 0.12. A total of 300 epochs were trained with a batch size of 8. During both training and testing, the input sizes of the images were set to 640 × 640, we refer to the literature [38] training set and test set division strategy, and choose 17 990 images/image pairs as the training set and 1469 images/image pairs as the test set. In order to prevent drastic oscillations at the start of training, we introduced the warm-up trick [64]. This allows the model to gradually stabilize under the low warm-up learning rate. For data augmentation, we used the light-aware mosaic data augmentation method, which mixes four training images based on their illumination intensity. Note that the processing of visible and thermal images should be

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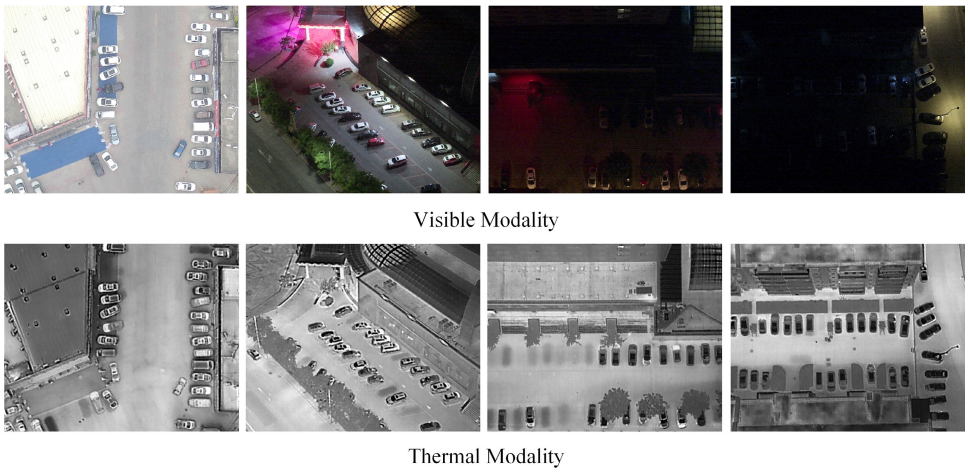


Fig. 13. Some examples of visible and thermal image pairs from the Drone Vehicle dataset.

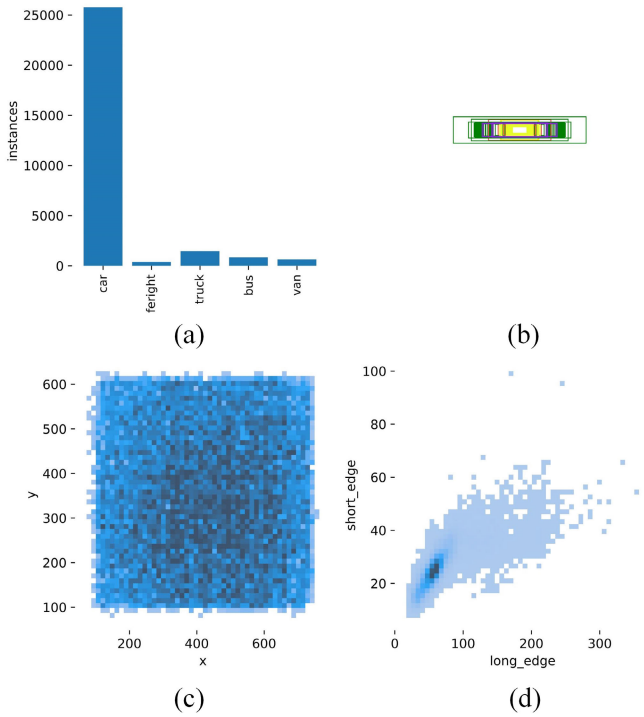


Fig. 14. Associated statistics of the Drone Vehicle dataset. (a) Number of objects of each type in the training set. (b) Size and number of the real bounding boxes of different categories in the training set. (c) Position of the center point of the object in relation to the whole image. (d) Long-to-short edge ratio of objects in an image relative to the whole image.

consistent when using data augmentation. Otherwise, inputting misaligned visible and thermal image pairs will cause the model training to fail. Fig. 15 shows four groups of image pairs that were enhanced by the light-aware mosaic data augmentation method. Experimental results for each group of our algorithms are averaged over ten training and testing runs under the same conditions. Compare algorithm training and testing strategies using their corresponding articles.

C. Evaluation Metrics

We use precision, recall, $AP_{0.5}$ (average precision), and $mAP_{0.5}$ (mean average precision) metrics to evaluate the performance of the CRSIOD algorithm and compare it with other methods, which are calculated as follows:

$$\text{Precision} = \frac{f_{\text{TP}}}{f_{\text{TP}} + f_{\text{FP}}} \tag{33}$$

$$\text{Recall} = \frac{f_{\text{TP}}}{f_{\text{TP}} + f_{\text{FN}}} \tag{34}$$

$$\begin{aligned} \text{AP} &= \int_0^1 \text{PrecisiondRecall} \\ &= \int_0^1 \frac{f_{\text{TP}}}{f_{\text{TP}} + f_{\text{FP}}} \text{d} \frac{f_{\text{TP}}}{f_{\text{TP}} + f_{\text{FN}}} = \int_0^1 P(r)\text{d}r \end{aligned} \tag{35}$$

$$\text{mAP} = \frac{1}{n} \sum_{i=0}^{n} \text{AP}_i = \frac{1}{n} \int_0^1 P_i(r)\text{d}r \tag{36}$$

where f_{TP} represents the cases where the model correctly identifies an object. f_{FP} represents the opposite, where the model incorrectly identifies an object that is not actually there. f_{FN} represents the cases where the model fails to detect an object that is actually there. Equation (35) shows that AP is the integral of the precision–recall curve (PRC) for each category. $mAP_{0.5}$ calculates the average of AP values for all categories at IoU $= 0.5$ in (36).

D. Ablation Study

The ablation experiments were performed on the Drone Vehicle dataset and will be analyzed in detail in this section.

1) Effectiveness of CSL-Based Angle Classification Loss: We choose the unimodal YOLOv5s as the baseline and improve it into a rotating object detection network using the angular regression-based method and the CSL-based method, respectively, and conduct experiments under the same conditions. The loss function of the angular regression-based method is selected as smooth $L1$ loss [13], and the experimental results are shown in Table II. The detection performance of the CSL-based method was 2.58% higher than that of the angular regression-based method (53.75% versus 51.17%). We can see from the visualized result graphs (Fig. 16) that the detection results of the angular regression-based approach show inaccurate prediction of multiple target angles (within the red ellipse). Experiments demonstrate the

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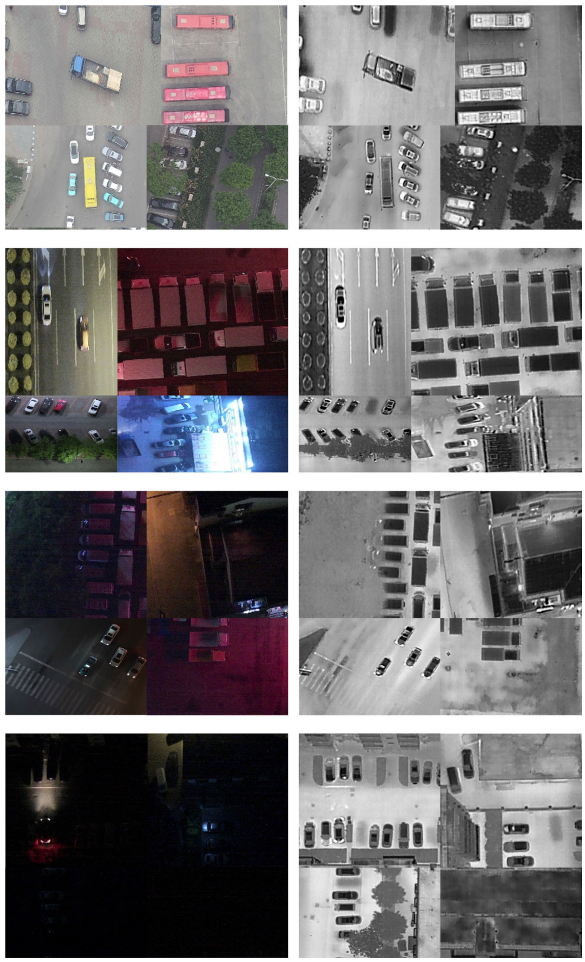
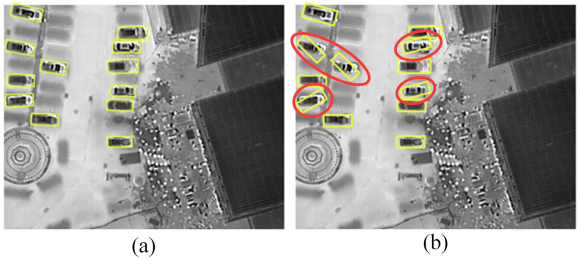


Fig. 15. Light-aware mosaic data augmentation sample.

TABLE II
EFFECT OF DIFFERENT ANGLE LOSS FUNCTIONS ON NETWORK DETECTION PERFORMANCE

Method	Angle loss	mAP
YOLOv5s(OBB)	regression-based	0.5117
	CSL-based	0.5375



(a) (b)

Fig. 16. Visualization of detection results with different angle loss functions applied to the network. (a) Detection results of the unimodal YOLOv5s network applying the CSL-based method as the angular loss function. (b) Detection result of applying the angular regression-based method as the angular loss function for the unimodal YOLOv5s network.

effectiveness of applying the CSL-based method as an angular loss function.

2) Impact of Fusion Stage and Fusion Method: To explore the impact of different fusion stages and fusion methods on network performance, we choose unimodal YOLOv5s as the baseline and improve it into a rotating box dual-stream object detection network. We use the original data augmentation method of YOLOv5s, the experiment was carried out under the same conditions, and the results are shown in Table III. We can see that the performance of the unimodal baseline algorithm in detecting visible images is 5.02% lower than the performance in detecting thermal images (48.73% versus 53.75%), and it is at least 6.32% (48.73% versus 55.05%) lower than the performance of the multimodal object detection algorithm, with a maximum difference of 9.50% (48.73% versus 58.23%). This is due to the large number of low-light or even no-light scenarios in the Drone Vehicle dataset, where the performance of the visible detector is minimal. The limitations of visible in ambient lighting conditions result in degraded image quality and difficulty in distinguishing the object from the background. In contrast, thermal image object detection algorithms and multimodal object detection algorithms perform better in these scenarios, providing more accurate object detection results. Rectangular unidentified objects with a shape similar to the silhouette of the target are common in thermal images, and vehicle-like shadows are also frequently seen in thermal images due to the effects of thermal imaging devices. All these factors have a significant impact on the performance of the thermal image object detection algorithm, resulting in a performance that is at least 1.30% (53.75% versus 55.05%) lower than the multimodal object detection algorithm, with a maximum difference of 4.48% (53.75% versus 58.23%). In the multimodal object detection algorithms Fusion-addI, Fusion-addII, Fusion-catI, and Fusion-catII, compared with the detection algorithms using the mid-term fusion methods (Fusion-addII and Fusion-catII), the detection performance using the early fusion methods (Fusion-addI and Fusion-catI) was reduced by at least 0.17% (55.05% versus 55.22%), with a maximum difference of up to 1.22% (55.09% versus 56.31%). This phenomenon is mainly caused by the following reason: during the early fusion process, thermal image features and visible image features not separately learned by the network are directly fused, thus introducing a large amount of redundant information. In addition, thermal and visible images reflect different effects on the target, with thermal image features focusing more on the target contours and visible image features focusing more on the texture details of the target. This difference leads to interference between the two modes, which in turn affects the learning ability of the detector. For mid-term fusion, the network first learns thermal image features and visible image features separately and then performs feature fusion after learning the target features of both modalities, which can alleviate the interference caused by redundant features of both modalities; therefore, our algorithm uses a mid-term fusion approach. Compared with the detection algorithm with the CMAFF fusion module, the detection algorithm with simple summation or concatenation fusion is at least 1.92% (56.31% versus 58.23%) worse in terms of detection performance, with a maximum difference of 3.01% (55.22% versus 58.23%). Traditional direct summing or concatenated fusion methods fail to efficiently extract the shared features and unique complementary features between the two modalities while

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TABLE III

EFFECT OF DIFFERENT FUSION STAGES AND FUSION METHODS ON NETWORK DETECTION PERFORMANCE. BASELINE IS THE SINGLE-MODE ROTATING BOX YOLOV5S (OBB), FUSION-ADDI IS THE EARLY FUSION BY ELEMENT SUMMATION, FUSION-ADDII IS THE MID-TERM FUSION BY ELEMENT SUMMATION, FUSION-CATI IS THE EARLY FUSION BY CONCATENATION, FUSION-CATII IS THE MID-TERM FUSION BY CONCATENATION, FUSION-CMAFF IS THE FUSION BY CMAFF MODULE, V IS THE VISIBLE MODE, AND T IS THERMAL MODE, AND LADE INDICATES THE USE OF THE LIGHT-AWARE MOSAIC DATA AUGMENTATION METHOD

Method	Modality	LADE	Car	Freight car	Truk	Bus	Van	mAP
Baseline	V	-	0.7016	0.1982	0.4861	0.8032	0.2475	0.4873
Baseline	T	-	0.8745	0.1724	0.5338	0.8532	0.2534	0.5375
Fusion-addI	V+T	-	0.8925	0.1714	0.5412	0.8646	0.2826	0.5505
Fusion-addI	V+T	✓	0.8922	0.2614	0.5436	0.8528	0.3012	0.5702
Fusion-addII	V+T	-	0.8958	0.1727	0.5448	0.8649	0.2869	0.5522
Fusion-addII	V+T	✓	0.8915	0.2633	0.5440	0.8598	0.3054	0.5728
Fusion-catI	V+T	-	0.8915	0.1801	0.5384	0.8625	0.2821	0.5509
Fusion-catI	V+T	✓	0.8915	0.2655	0.5384	0.8625	0.3221	0.5760
Fusion-catII	V+T	-	0.9078	0.2098	0.5439	0.8685	0.2856	0.5631
Fusion-catII	V+T	✓	0.8989	0.2909	0.5419	0.8655	0.3356	0.5866
Fusion-cmaff	V+T	-	**0.9182**	0.2146	**0.5689**	**0.8882**	0.3218	0.5823
Fusion-cmaff	V+T	✓	0.9181	**0.3402**	0.5611	0.8792	**0.3618**	**0.6121**

TABLE IV

EFFECT OF UAM AND LP-NMS ON THE MODEL

Method	UAM	LP-NMS	mAP
Fusion-cmaff	-	-	0.6121
Fusion-cmaffI	✓	-	0.6305
Fusion-cmaffII	-	✓	0.6286
Fusion-cmaffIII	✓	✓	**0.6451**

TABLE V

COMPARISON OF MODEL DETECTION ABILITY UNDER DIFFERENT ATTENTION MECHANISMS. *r* IS THE SHRINKAGE RATE

Attention mechanism	r	mAP
-	-	0.6451
SE	8	0.6490
	16	0.6509
	32	**0.6606**
CBAM	8	0.6515
	16	0.6560
	32	**0.6590**
CA	8	0.6755
	16	0.6703
	32	**0.6846**

introducing interference features. The CMAFF fusion module can effectively enhance complementary features between modalities and extract common features of modalities, thus reducing the effect of redundant features.

3) Impact of the Light-Aware Mosaic Data Augmentation: In order to verify the effectiveness of the light-aware mosaic data augmentation, we trained five multimodal detection networks with the light-aware mosaic data augmentation. We can see from the detection results in the LADE drawing ✓ section of Table III that the accuracy of all five multimodal detection networks showed significant improvement in accuracy. The Fusion-add I network improved by 1.97% (55.05% → 57.02%), while Fusion-add II improved by 2.06% (55.22% → 57.28%). Fusion-cat I showed an improvement of 2.51% (55.09% → 57.60%), followed by Fusion-cat II with a 2.35% (56.31% → 58.66%) improvement. Finally, Fusion-cmaff achieved the highest improvement of 2.98% (58.23% → 61.21%). This is because the images generated using the light-aware mosaic data augmentation approach are closer to the real scene, and the network can effectively learn the scene features and target features.

4) Impact of the UAM and LP-NMS: Table IV shows the effect of the two modules, uncertainty-aware module (UAM) and LP-NMS, on the detection model when acting separately and together. In the daytime, we set W_{vis} to 1 because visible image targets in the daytime have clear texture and detail information, thermal images may be negatively affected by thermal radiation from nonreal targets with high temperatures, so the visible image annotation results are more accurate compared with thermal images, so it is used as a reference model to compensate for missing targets in thermal images. At night, we take different values of W_{vis}, ranging from 0.01 to 0.5, for the experiments. Experimental results show that the model achieves the highest mAP when W_{vis} is set to 0.1. We take different values of W_{the} from 0.01 to 0.5 for the experiments during the day. Experimental results show that the model achieves the highest mAP, when W_{the} is set to 0.1. We set the setting to 1 during the night. Since the thermal image is not affected by light and the labeling result is more accurate, it is used as a reference model to compensate for missing targets in the visible image. Two modal target alignment thresholds μ was set to 0.8 with reference to the literature [38]. For Fusion-CMAFFI, only the UAM module was used, resulting in a 1.84% (61.21% → 63.05%) increase in mAP. This is due to the fact that the UAM module can determine the network to learn in a direction favorable to object detection, and during the daytime, the UAM prompts the

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TABLE VI
QUANTITATIVE COMPARISON BETWEEN THE PROPOSED METHOD AND OTHER METHODS

Method	Modality	Car	Freight car	Truk	Bus	Van	mAP
RetinaNet [17]	V	0.6750	0.1372	0.2824	0.6205	0.1926	0.3816
	T	0.7986	0.2805	0.3284	0.6732	0.1644	0.4490
Faster R-CNN [65]	V	0.6788	0.2631	0.3859	0.6698	0.2320	0.4459
	T	0.8863	0.3516	0.4251	0.7792	0.2852	0.5455
Faster R-CNN(Dpool) [66]	V	0.6823	0.2640	0.3873	0.6908	0.2638	0.4576
	T	0.8894	0.3679	0.4791	0.7828	0.3279	0.5694
Mask R-CNN [67]	V	0.6852	0.2683	0.3984	0.6675	0.2523	0.4546
	T	0.8877	0.3663	0.4886	0.7838	0.3216	0.5696
Cascade Mask R-CNN [14]	V	0.6800	0.2725	0.4467	0.6934	0.2980	0.4781
	T	0.8100	0.3897	0.4718	0.7932	0.3300	0.5589
Hybrid Task Cascade [68]	V	0.6789	0.2722	0.4455	0.7022	0.2861	0.4770
	T	0.8857	0.4285	0.4771	0.7946	0.3416	0.5855
RoITransformer [21]	V	0.6813	0.2908	0.4417	0.7055	0.2764	0.4791
	T	0.8885	0.4149	0.5153	0.7948	0.3439	0.5915
ReDet [69]	V	0.6948	0.3146	0.4787	0.7737	0.2903	0.5104
	T	0.8847	0.4282	0.5395	0.7989	0.3656	0.6054
Gliding Vertex [26]	V	0.7577	0.3375	0.4608	0.6805	0.3872	0.5248
	T	0.8915	0.4295	0.5972	0.7875	0.4388	**0.6289**
Yolov5s(OBB)	V	0.7016	0.1982	0.4861	0.8032	0.2475	0.4873
	T	0.8745	0.1724	0.5338	0.8532	0.2534	0.5375
UACMD [38]	V+T	0.8751	**0.4680**	0.6070	0.8708	0.3795	0.6401
Fusion-addI	V+T	0.8925	0.1714	0.5412	0.8646	0.2826	0.5505
Fusion-addII	V+T	0.8958	0.1727	0.5448	0.8649	0.2869	0.5522
Fusion-catI	V+T	0.8915	0.1801	0.5384	0.8625	0.2821	0.5509
Fusion-catII	V+T	0.9078	0.2098	0.5439	0.8685	0.2856	0.5631
Fusion-cmaff	V+T	0.9182	0.2146	0.5689	0.8882	0.3218	0.5823
CRSIOD	V+T	**0.9458**	0.4629	**0.6367**	**0.9202**	**0.4574**	**0.6846**

TABLE VII
COMPARISON OF MODEL DETECTION ABILITY ON THE LLVIP DATASET

Methods	Modality	mAP
Faster R-CNN [13]	V	0.8895
	T	0.9373
YOLOv3 [46]	V	0.8586
	T	0.8965
YOLOv5 [48]	V	0.9081
	T	0.9463
CFT [70]	V+T	0.9751
UACMD [38]	V+T	0.9630
Fusion-addI	V+T	0.9480
Fusion-addII	V+T	0.9510
Fusion-catI	V+T	0.9487
Fusion-catII	V+T	0.9589
Fusion-cmaff	V+T	0.9638
CRSIOD	V+T	**0.9810**

network to take full advantage of the daytime visible images by giving the visible image detection branch a larger weight to make it dominant, and assigning the thermal image detection branch a smaller weight so that it can assist the visible image detection branch as well as avoiding the interference of certain nontarget objects that have a high temperature in the daytime. At night, UAM prompts the network to give full play to the advantages of thermal images, giving the thermal image detection branch a larger weight to make it dominant, and assigning the visible image detection branch a smaller weight so that it can not only assist the thermal image detection branch but also to avoid the negative impact of the visible image in low light. For Fusion-CMAFFII, only the LP-NMS module was used, leading to a 1.65% (61.21% $\rightarrow$ 62.86%) increase in mAP. This is because visible images are difficult to accurately predict real objects in dark scenes. When the network reaches the final stage to synthesize the predictions of the three branches, the false positive samples predicted by the visible detection branch severely affect the final detection performance. Such situations mainly occur in night scenes, so we use the illumination perception module to guide the NMS to reduce the interference of false positive samples generated by the visible image detection branch. For Fusion-CMAFFIII, both the UAM and LP-NMS modules were applied to the detection model, resulting in a 3.30% (61.21% $\rightarrow$ 64.51%) increase in mAP. These experiments demonstrate the effectiveness of the UAM and LP-NMS modules.

5) Impact of the Attentional Mechanisms: Different attention mechanisms were imposed on Fusion-CMAFFIII, and Table V lists the comparison of the effect of imposing different attention mechanisms in the network on the detection capability of the network. Utilizing the SE attention mechanism with a shrinkage rate r of 32 improved the detection module performance to 66.06%, a 1.55% enhancement (64.51% $\rightarrow$ 66.06%).

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TABLE VIII
COMPARISON OF MODEL DETECTION ABILITY ON THE M^{3}FD DATASET

Methods	Modality	People	Car	Bus	Lamp	Motorcycle	Truck	mAP
Faster R-CNN [13]	V	0.6499	0.8285	0.8378	0.7526	0.6500	0.7611	0.7467
	T	0.6870	0.8311	0.8401	0.7635	0.6512	0.7743	0.7579
YOLOv3 [46]	V	0.6009	0.8032	0.8098	0.6917	0.6188	0.7020	0.7044
	T	0.6300	0.8275	0.8336	0.7421	0.6233	0.7501	0.7344
YOLOv5 [48]	V	0.6541	0.8432	0.8560	0.7610	0.6432	0.7910	0.7581
	T	0.7071	0.8890	0.8615	0.7564	0.6548	0.8120	0.7801
CFT [70]	V+T	0.7183	0.9151	0.9245	**0.8170**	0.7082	0.8143	0.8162
UACMD [38]	V+T	**0.8021**	0.8902	0.8660	0.7033	0.7260	0.8284	0.8027
Fusion-addI	V+T	0.6985	0.8892	0.8654	0.7611	0.6618	0.8122	0.7814
Fusion-addII	V+T	0.7150	0.8896	0.8761	0.7570	0.6848	0.8133	0.7893
Fusion-catI	V+T	0.7100	0.8911	0.8702	0.7622	0.6621	0.8115	0.7845
Fusion-catII	V+T	0.7166	0.8901	0.8865	0.7689	0.6923	0.8038	0.7930
Fusion-cmaff	V+T	0.7531	0.9012	0.8886	0.7711	0.7013	0.8200	0.8059
CRSIOD	V+T	0.7840	**0.9222**	**0.9326**	0.8046	**0.7387**	**0.8553**	**0.8396**

By applying the CBAM attention mechanism with a shrinkage rate r of 32, the detection module achieved the highest performance at 65.90%, demonstrating a 1.39% improvement (64.51% $\rightarrow$ 65.90%). The application of the CA attention mechanism with a shrinkage rate r of 32 resulted in the highest performance (68.46%) for the detection module, showing a substantial improvement of 3.95% (64.51% $\rightarrow$ 68.46%). r represents the shrinkage ratio of the attention block. Through experiments, we found that it is not better to apply the attention mechanism to the network with a larger shrinkage rate or a smaller shrinkage rate, which needs to be based on its own research task to carry out multiple experiments to select the appropriate shrinkage rate to achieve the best performance. For the task in this article, we found that the three attention mechanisms achieve the best performance with a shrinkage rate of r of 32, which is due to the fact that the target is too dense and the target characteristics are not prominent in the research task in this article. Applying more fine-grained attention to the scene will increase the redundant information (especially in the thermal image, there are a lot of unidentified objects that are similar to the target), which not only increases the computational complexity but also reduces the detection performance of the algorithm.

E. Qualitative Analysis

As shown in Fig. 17, we qualitatively compare CRSIOD with the real annotated image and the best performing multimodal detection network Fusion-catII under different lighting conditions and different complex scenes. Visually, our CRSIOD rarely has false detections (green arrows) or misses (red arrows). It performs better in detecting difficult samples such as compact small targets and heavily occluded targets. In contrast, Fusion-catII has multiple misses, especially for densely distributed targets, which are seriously neglected. Moreover, CRSIOD also learns the features of strongly occluded targets and is still able to detect them accurately, even if they were manually missed (blue arrows). Although this may result in some loss of accuracy, it has significant value in engineering practice and automated labeling work.

F. Comparison With State-of-the-Art Methods

Table VI shows the detection results of existing methods and our model on the Drone Vehicle dataset. It can be seen that our CRSIOD method performs better on this dataset. Our CRSIOD method has a 5.57% higher mAP compared with the Gliding Vertex algorithm with the best detection performance of the unimodal detection network (62.89% versus 68.46%). Our CRSIOD method has 4.45% higher mAP compared with the UACMD algorithm, which has the best detection performance of the multimodal detection network (64.01% versus 68.46%). It should be noted that our CRSIOD ($\sim$18.26M) is a lightweight network with far fewer parameters than UACMD ($\sim$138.69M).

G. Generalization to Multimodal Complex Scene Images

Despite the existence of numerous publicly available aerial image datasets, most of them are limited to a single data mode. To validate the generalization performance of our proposed network, we collect two datasets containing complex nighttime scenes, aiming to demonstrate its ability to handle such scenarios with high generalization capability. These datasets include the visible–thermal pairing dataset LLVIP [39] for pedestrian detection in low-light scenes and the visible–thermal pairing dataset M^{3}FD [40] for multicategory detection in complex scenes.

1) LLVIP: LLVIP is the latest release of the visible–thermal paired pedestrian dataset for low-light vision. The dataset collected 15 488 pairs of visible–thermal images from 26 different locations, most of which were acquired in extremely dark environments. Some examples of visible and thermal image pairs in the LLVIP dataset are shown in Fig. 18. In our experiments, we refer to the literature [39] training set and test set division strategy, and choose 12 025 images/image pairs as the training set, 3463 images/image pairs as the test set, and the image size is fixed at 640 $\times$ 640.

2) M^{3}FD: The M^{3}FD dataset comprises a total of 4200 pairs of visible–thermal images, categorized into four distinct scene types: daytime, overcast, nighttime, and

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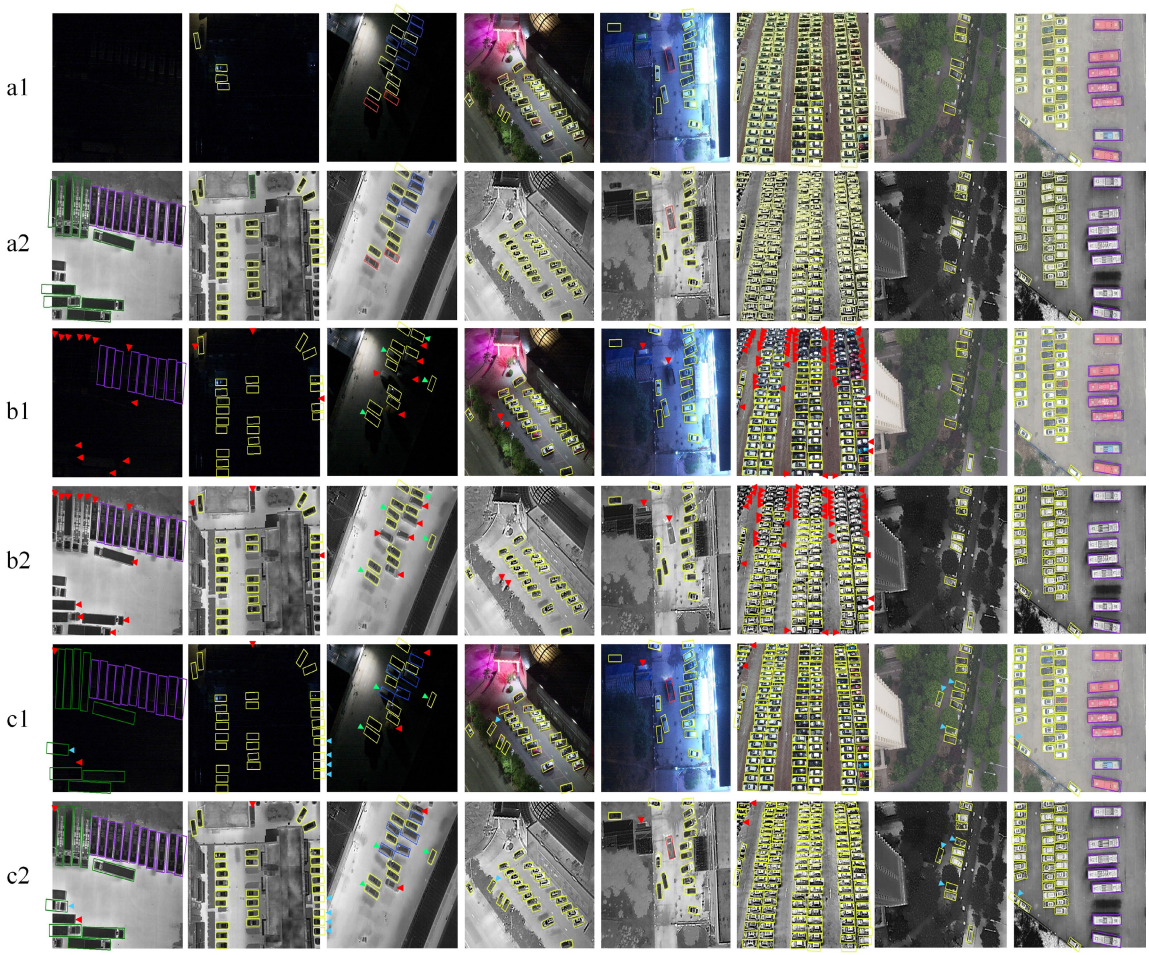


Fig. 17. Visualization of detection results (a1) and (a2) visible and thermal real labeled images, respectively, (b1) and (b2) visible and thermal images detected by Fusion-catII, respectively, and (c1) and (c2) visible and thermal images detected by CRSIOD, respectively.



Fig. 18. Some examples of visible and thermal image pairs from the LLVIP dataset.

challenging conditions. Within this dataset, 33 603 objects belonging to six different classes have been annotated, including people, car, bus, motorcycle, truck, and lamp. Some examples of visible and thermal image pairs in the M^3FD dataset are shown in Fig. 19. In our experiments, we choose 3780 images/image pairs as the training set, 420 images/image pairs as the test set, and the image size is fixed at 640 × 640.

The associated statistics are shown in Fig. 20 for the LLVIP dataset and in Fig. 21 for the M^3FD dataset. Both datasets are labeled with horizontal bounding boxes; hence, angular classification layers with angular classification loss are removed from the CRSIOD head network.

In order to verify the generalization of the CRSIOD algorithm proposed in this article, we select three unimodal detection algorithms, Faster R-CNN [13], YOLOv3 [46], YOLOv5 [48] and seven multimodal detection algorithms, CFT [70], UACMD [38], Fusion-addI, Fusion-addII, Fusion-catI. Fusion-catII, and Fusion-CMAFF.

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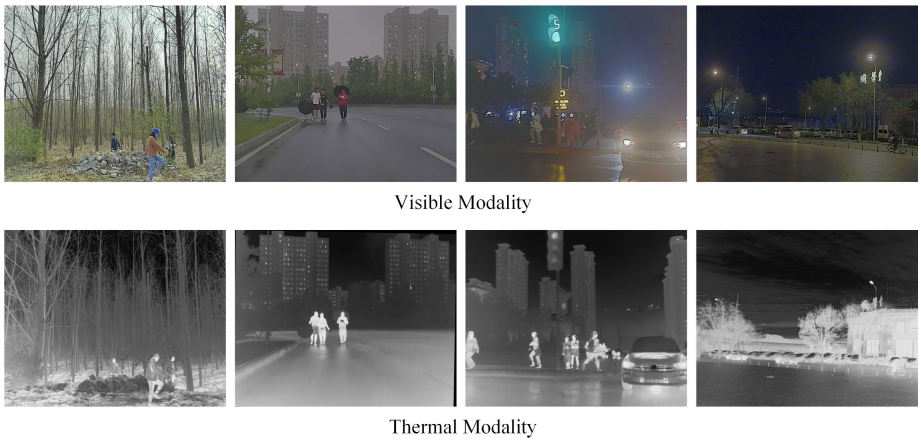


Fig. 19. Some examples of visible and thermal image pairs from the M^{3}FD dataset.

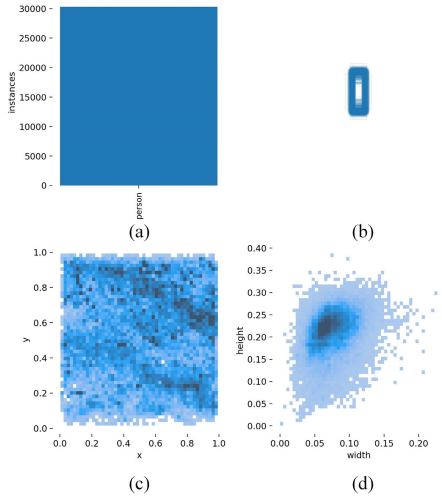


Fig. 20. Associated statistics of the LLVIP dataset. (a) Number of objects of each type in the training set. (b) Size and number of the real bounding boxes of different categories in the training set. (c) Position of the center point of the object in relation to the whole image. (d) Aspect ratio of objects in an image relative to the whole image.

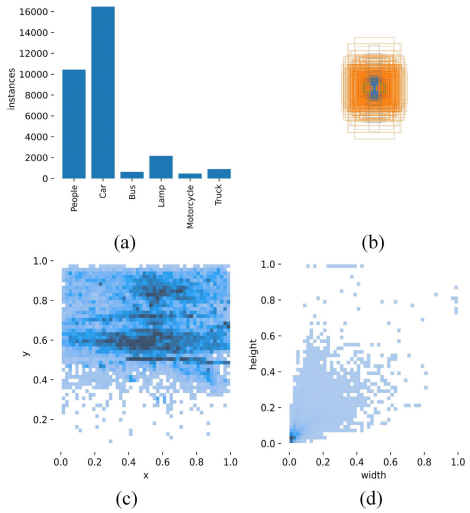


Fig. 21. Associated statistics of the M^{3}FD dataset. (a) Number of objects of each type in the training set. (b) Size and number of the real bounding boxes of different categories in the training set. (c) Position of the center point of the object in relation to the whole image. (d) Aspect ratio of objects in an image relative to the whole image.

From Tables VII and VIII, we can find that for the unimodal object detection algorithms, the performance of the object detection algorithm in visible mode is at least 3.79% (85.86% versus 89.65%) and 1.12% (74.67% versus 75.79%) lower than that of the object detection algorithm in thermal mode, with a maximum difference of 4.78% (88.95% versus 93.73%) and 3.00% (70.44% versus 73.44%). This is due to the inherent problem of visible modes, where the target is affected by illumination and occlusion, as shown in Figs. 18 and 19. First, the visible modality is affected by illumination conditions, where the intensity and direction of the light may cause changes in the appearance of the target, shadow creation, and loss of detail. Such lighting variations make it more difficult to accurately detect targets in visible images and can lead to performance degradation. Second, occlusion problems in visible images can also negatively affect object detection performance. When a target is occluded by other objects, the boundaries and features of the target may be partially or completely hidden, making it difficult for the detector to accurately locate and identify the target.

For thermal image detectors, despite their better performance compared to visible image detectors, there are still some issues, such as device noise and imperfect scene information. First, thermal images may be affected by noise from the device itself. This may interfere with the ability of object detection algorithms to accurately locate and identify objects in images. The presence of noise may lead to false or missed detections of the target, reducing the performance of the detector. Second, since thermal images reflect the thermal radiation of an object, unlike visible images, scene information in thermal images is usually not as rich as in visible images. This can result in some object-like shapes being misidentified as targets, or some real targets being missed due to imperfect information in the thermal images.

In addition, a target in a thermal image may also be missed when the obscuring object is similar to the thermal radiation of the target itself. The performance of multimodal object detection is significantly higher than that of unimodal object detection. This suggests that multimodal object detection algorithms have more comprehensive sensing capabilities and

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can leverage multiple modal information simultaneously to improve the accuracy and robustness of object detection. On the LLVIP dataset, the CFT algorithm has only a 0.59% (97.51% versus 98.10%) performance difference with respect to the CRSIOD algorithm, while on the M^3FD dataset, the CFT algorithm has a lower performance with respect to the CRSIOD algorithm by 2.34% (81.62% versus 83.96%). This is due to the fact that the M^3FD dataset contains more complex scenes compared with the LLVIP dataset, which suggests the high accuracy and robustness of our CRSIOD algorithm in the complex scene object detection task. It is worth noting that the CFT algorithm has a parameter count of (∼206.27M) and computational complexity that is 11.30 times higher than that of the CRSIOD algorithm (∼18.26); thus, our model has a better balance in terms of considering detection accuracy and speed.

V. Conclusion

In this work, we proposed a new network called CRSIOD. First, we designed the illumination perception module, which guides the object detection network to perform various feature processing tasks by judging the illumination conditions in the visible image scene. Second, we designed the uncertainty-aware module, which converts various uncertainties between the two modalities into weight values to be applied to the object detection network, so that the feature learning of the network is carried out in the direction favorable to object detection, which greatly brings into play the advantages of the two modalities themselves and circumvents their respective negative effects. In addition, in the object detection network, the dual-stream backbone network based on the attention mechanism plays an important role in the learning of difficult samples, which enhances the target features of the two modalities to inhibit the interference of the background, and the fusion modal features obtained by using the CMAFF module fully retains the shared features and complementary features of the two modalities. We feed the three modalities into the feature enhancement network to fully exploit the unique feature advantages of each modality. In the final result fusion part, the light-aware LP-NMS reduces the interference of false positive samples generated by the visible image detection branch and improves the detection performance of the object detection network, and for the Drone Vehicle dataset, the horizontal frame detector head is improved to a rotating frame detector head, which retains the directionality of the target and reduces the redundant information. A large number of experiments prove that our CRSIOD method can effectively utilize the respective features of visible and thermal images and has better detection capability under different illumination environments and different complex scenes, reflecting the characteristics of high accuracy and robustness, especially our CRSIOD fully learns the strong occlusion target features and is still able to detect the targets with strong occlusion for manual miss labeling, which is engineering practice and automated annotation work has important application value. Compared with the proposed UAV aerial image detection methods in recent years, our proposed CRSIOD method performs well.

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